

● EASY

● ADVANCED MATH

● ANSWER KEY

Advanced Math

15

3 answers · Rationale-rich · Teach from doc

Q1**C****C.** 17

Choice C is correct. It's given that $fx = x^3 + 9$. Substituting 2 for x in this equation yields $f2 = 2^3 + 9$. This is equivalent to $f2 = 8 + 9$, or $f2 = 17$.

Choice A is incorrect. This is the value of $2 + 3 + 9$, not $2^3 + 9$.

Choice B is incorrect. This is the value of $23 + 9$, not $2^3 + 9$.

Choice D is incorrect. This is the value of $3^2 + 9$, not $2^3 + 9$.

Q2**A****A.** 118

Choice A is correct. Substituting 2 for x in the given equation yields $k(2) = 2^3 + 110$, which is equivalent to $k(2) = 8 + 110$, or $k(2) = 118$. Therefore, the value of $k(x)$ when $x = 2$ is 118.

Choice B is incorrect. This is the value of $k(x)$ when $x = 2$ if $k(x) = 3x + 110$, not $k(x) = x^3 + 110$.

Choice C is incorrect. This is the value of $k(x)$ when $x = 2$ if $k(x) = x + 3 + 110$, not $k(x) = x^3 + 110$.

Choice D is incorrect. This is the value of $k(x)$ when $x = 0$, not $x = 2$.

Q3

B

B. The marble's height was 8 inches above the ground when it started moving.

Choice B is correct. The y -intercept of a graph is the point at which the graph intersects the y -axis. The graph shown intersects the y -axis at the point $(0, 8)$. Therefore, the y -intercept of the graph is $(0, 8)$. It's given that y is the height of the marble above the ground, in inches, and x is the number of seconds after the marble started moving. It follows that the marble's height was 8 inches above the ground 0 seconds after it started moving. Therefore, the best interpretation of the y -intercept of the graph is that the marble's height was 8 inches above the ground when it started moving.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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